

Department of Computer Science & Engineering
Punjabi University Patiala

MST: 1

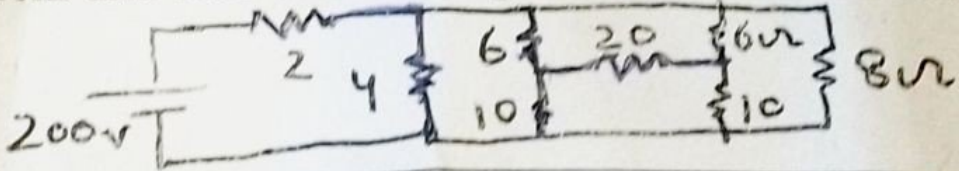
Subject: Basic Electrical Engineering

Max. Mark: 15

Section: A (All questions are compulsory)

Q1. (A)	State Kirchhoff's laws?	1
Q1.(B)	What is Form factor and power factor?	1
Q1.(C)	Define Q-factor of series resonance?	1
Q1. (D)	Transform star to delta $R_1=5$ ohm, $R_2=10$ ohm, $R_3=20$ ohm.	1
Q1. (E)	Define maximum power transfer theorem?	1

Section: B (Do any two Questions)

Q2. (A)	Explain RC series circuit with phasor diagram?	2.5
Q2. (B)	An ac series circuit has a $R=10$ ohm, $L=0.2H$ and $C=60\mu F$ Calculate 1) resonant frequency, 2) current, 3) power at resonance Give that the applied voltage is 200V.	2.5
Q3.	State Norton's theorem and find current using Norton's theorem through a 8 ohm in the given circuit. 	2 3
Q4.	State Thevenin theorem. Elaborate it step by step.	5

Department of Computer Science Engineering
(Punjabi University, Patiala)

Subject: Basic Electrical Engineering

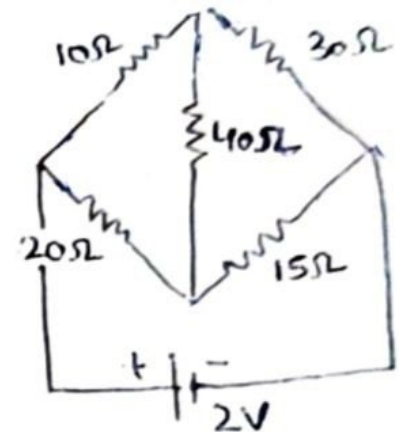
Time: 1 hour
Max. Marks: 15

(SECTION-A) (Attempt all questions) (1 × 5)

1. (a) Write the equivalent resistance between AB in ABC delta connection. Each resistance value is R.
(f) Define Form Factor.
(g) State Maximum Power transfer theorem.
(h) What is Impedance?
(i) Define Q-Factor in Parallel resonance.

(SECTION-B) Attempt any Two (2 × 5)

Q2. A bridge network ABCD is arranged as follows; resistance between terminals A-B, B-C, C-D, D-A and B-D are 10, 30, 15, 20 and 40 ohm respectively. A 2V battery is connected across terminals A and C. Determine the value and direction of the current in 40 ohm resistor by applying Thevenin theorem.



Q3. A coil of resistance 10 ohms and inductance 0.1H is connected in series with a condenser of capacitance 150 micro farad across a 200V, 50Hz supply. Determine (i) impedance (ii) current (iii) power factor (iv) voltage across coil (v) voltage across condenser.

Q4. Discuss the behavior of RLC series circuit when applied with ac voltage with the help of phasor diagram. Also, derive the expression for series resonance.